overlap. To solve such a situation, Ss$^+$-trees introduce two new elements:

1. A check to discover such situations;
2. A new heuristic that applies k-means to all grand-children of a node `N` (no matter if they are points or other nodes, in the latter case their centroids will be used for clustering), and the clusters created will replace `N`'s children.

To check the overlap, the first thing we need is the formula to compute the volume of intersections of two spheres: unfortunately computing the exact overlap of two hyperspheres, in the generic k-dimensional case, requires not just substantial work and good calculus skills to derive the formulas, but also robust computing resources, as it resulting in a formula that includes an integral, whose computation is clearly expensive enough to question its usefulness in a heuristic.

An alternative approach is to check if one of the two bounding envelops is completely included in the other: checking that the center of one sphere is contained in the other one, and that the distance of the centroid of the smaller sphere is closer that `R-r` to the centroid of the larger one, where `R` is the radius of the larger sphere and, as you might expect, `r` is the radius of the smaller one.

Variants of this check can set a threshold for ratio between the actual distance of the two centroids and `R-r`, using it for an approximation of the overlapping volume as this ratio gets closer to `1`.